



Republic of the Philippines
Department of Agriculture
BUREAU OF FISHERIES AND AQUATIC RESOURCES
REGIONAL FISHERIES LABORATORY XII
J. Catolico St., Lagao, General Santos City

Document Type Work Instruction	Revision No: 0	Date Adopted: 13 Aug 2019	Page No: Page 1 of 7
Document Code: WI-01-CHE-02	HISTAMINE		

1. PURPOSE

To provide a standardized method for the analysis of histamine using the fluorometric method of analysis

2. SCOPE

This procedure should be followed in the determination of histamine in fish and fishery products.

3. RESPONSIBILITY

The analyst performing the test should ensure that the procedure is followed accordingly.

4. APPARATUS

- 4.1 Beaker – 50, 100 and 250 ml
- 4.2 Filter paper Whatman No. 1
- 4.3 Volumetric Flask – (10, 25, 50, 100, 500 and 1000 ml)
- 4.4 Cotton/Glass wool
- 4.5 Volumetric Pipette – (1,2,3,4,5, 10 and 20 ml)
- 4.6 Graduated cylinder – 50 and 100 ml
- 4.7 Test tube with cover
- 4.8 Polypropylene tube 50 ml
- 4.9 Glass bottle 50 ml
- 4.10 Glass column
- 4.11 Glass funnel
- 4.12 pH paper 6-8
- 4.13 Mechanical pipette (1000 µl)

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5. EQUIPMENT

- 5.1 Homogenizer
- 5.2 Electronic balance
- 5.3 Water bath set at 60°C
- 5.4 Fluorometer
- 5.5 Alarm clock

6. REAGENTS

- 6.1 All information regarding reagent preparation is placed in LF-W04-01, Reagent and Media Preparation Logbook.
- 6.2 **Ion-exchange resin- BIO Rad AG 1-x8, 50-100 mesh or DOWEX 1-x8, 50-100 mesh**
 - 1. Convert to -OH form by adding 15 ml 2N NaOH/g resin to beaker. Swirl mixture and let stand for <30 min. Decant liquid and repeat with additional base. Thoroughly wash resin with H₂O, slurry into filter paper and wash again with H₂O until the wash water is neutral to pH paper, then store the prepared resin under water. Prepare fresh weekly.
 - 2. Prepare the ion exchange column by placing a cotton/glass wool plug in the base of the column, then slurry in sufficient resin to form an 8 cm deep bed, avoiding the formation of air-bubbles in the packed resin. Keep the resin covered with H₂O at all times. Wash column with 10 ml H₂O before applying next sample.
- 6.2 **3.57N Phosphoric acid** - Pipette 12.18 ml of 85% H₃PO₄ and make up the volume to 100 ml with deionized water or 121.8 ml for 1000 ml.
- 6.3 **0.1% O-phthaldialdehyde (OPT)** - Dissolve 0.1 g of OPT in 100 ml methanol. Store in amber bottle in refrigerator. Prepare weekly.

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6.4 **1N HCl** - Pipette 8.33 ml of conc. HCl and make up to 100 ml with deionized water or 83.3 ml for 1000 ml.

6.5 **0.1N HCl** - Pipette 8.33 ml of concentrated HCl and make up to 1L with deionized water.

6.6 **1N NaOH** - Dissolve 4g of NaOH in 100 ml of deionized water or 40 g in 1000 ml.

6.7 **2N NaOH** - Dissolve 8g of NaOH in 100 ml of deionized water or 80 g in 1000 ml.

6.8 **Histamine standard stock solutions (1000 ppm)**

Dissolve 0.1674g of standard histamine dihydrochloride (99%) in 100 ml of 0.1 N HCl. Store in amber bottle in refrigerator. Prepare weekly.

6.9 **Intermediate and working standard solution at various concentration. Diluent: 0.1N HCl**

Wanted concentration (ppm)	Conc. Of stock standard (ppm)	Volume of aliquot (mL)	Total volume (mL)
100	1000	10	100
10	100	10	100
1	10	10	100
0.3	1	30	100
0.2	1	20	100
0.1	1	10	100
0.05	0.1	50	100
0.01	0.1	10	100

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7. PROCEDURES

7.1 SAMPLE PREPARATION

7.1.1 Refer to WI-01-CHE-01

7.2 Quality Control: Preparation of Reagent Blank and Fortified Sample

7.2.1 Reagent Blank

7.2.1.1. For reagent blank, add 40 ml of methanol into 50 ml polypropylene tube

7.2.1.2. Proceed to step 7.3.5.

7.2.2 Spiked sample

7.2.2.1. Weigh 5.0 g (accuracy of 0.001 g) sample into a PP vessel.

7.2.2.2. Add 50ul of 1000 ppm histamine standard (expected recovery: 10 ppm)

7.2.2.3 Proceed as in 7.3.2.

7.3 SAMPLE EXTRACTION

7.3.1. Weigh 5 grams of sample into 50 ml polypropylene tube.

7.3.2. Add 20 ml of methanol.

7.3.3 Homogenize 1 minute

7.3.4 Add another 20 ml of methanol.

7.3.5 Place in water bath at 60°C for 15 mins.

7.3.6 Cool to room temperature.

7.3.7 Transfer sample mixture through glass funnel into a 50 ml volumetric flask.

7.3.8 Make up volume to 50 ml with methanol and shake.

7.3.9 Filter the mixture through Whatman No. 1 filter paper.

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Note: Filtrate can be stored for 1 month.

7.4 COLUMN CHROMATOGRAPHY ELUTION

- 7.4.1 Pass 10 ml of water through the ion exchange column and discard this solution.
- 7.4.2 Pipette 1 ml of the sample extract into a glass column containing an ion-exchange resin.
- 7.4.3 Add 5 ml of water. Immediately initiate the column flow and collect the eluate in a 50 ml volumetric flask which contains 5 ml of 1N HCl. Control flow rate at >3 ml/min.
- 7.4.4 When the liquid level is approximately 2 mm above the top of the resin, add water until the eluate is nearly 50 ml. Stop the flow and make up volume with water and mix.

7.5 CALIBRATION CURVE

- 7.5.1 Transfer 5 ml of 0.01, 0.05, 0.10, 0.20 and 0.30 ppm standard solution into five separate containers.
- 7.5.2 For calibration curve blank solution, use 5 ml of 0.1N HCl.
- 7.5.3 Continue from Step 7.7.1

7.6 CONTINUING CALIBRATION VERIFICATION

- 7.6.1. Prepare 0.1 ppm Histamine solution from standard
- 7.6.2. Proceed as in 7.7.1

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7.7 DERIVATIZATION AND FLUOROMETRIC READING

- 7.7.1 Pipette 5 ml of the calibration standard/column chromatography eluate /spike/ccv solution into a 50 ml container.
- 7.7.2 Add 10 ml of 0.1N HCl
- 7.7.3 Add 3 ml of 1N NaOH and mix
- 7.7.4 Within 5 minutes add 1 ml of 0.1% OPA solution and mix.
- 7.7.5 Just after 4 mins, add 3 ml of 3.57N H₃PO₄ and mix.
- 7.7.6 Within 1.5 hour, record fluorescence intensity (I) using excitation wavelength of 450 nm.
- 7.7.7 If fluorescence intensity of the sample is greater than the fluorescence intensity of maximum standard solution, the column chromatography eluate solution should be diluted with 0.1N HCl in the proper ratio.

8 CALCULATION

- 8.7 Plot of intensity, I (measured by fluorometer and corrected for blank) against ug histamine/5 ml test solution should be a straight line
- 8.8 The histamine concentration can be calculated as follows:

$$\text{Histamine (ppm)} = (C \times E \times 50 \times 50) / (D \times 1 \times W)$$

Where

C = amount of histamine from calibration curve (ug)

W = Weight of sample (g)

D = eluate solution volume for dilution (ml)

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E = total solution volume after dilution (ml)

9 REFERENCE:

Manual of On-site Training on Histamine Analysis (Trust Fund IV Project) organized by SEAFDEC, 25-30 September 2006 with reference to AOAC 17th ed. Revision 2, 2003. Chapter 35. Fish and other marine products. 35.1.32. p.17-19

10 FORMS

LF W01-01, Reagent and Media Preparation Logbook

LF W01-CHE-01, Analyst Worksheet for Histamine Analysis

WI-01-CHE-01, Sample Preparation (Meat Samples)

11 DOCUMENT HISTORY

Revision No.	Issue Date	Original Entry	Revised as	Recall Date

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